Lesson 3.1 Guided Notes

Session 1 · Understanding Ratios

I can describe a relationship between two quantities using ratio language. · 6.RP.A.1

Word / Spanish	Meaning — complete the blank	Example
ratio / razón	A comparison of two _____.	3 to 5, 3 : 5, 3/5
term / término	One of the two _____ in a ratio.	In 3 : 5, the terms are 3 and 5.
part-to-part / parte a parte	Compares one group with _____ group.	apples : oranges
part-to-whole / parte al todo	Compares one group with the _____.	apples : all the fruit

Key idea

Write a ratio in the _____ the question asks.

For part-to-whole, _____ the groups to find the whole.

To simplify, divide both terms by the same _____.

Worked example

A shelf has 6 math books and 9 reading books. Write math to all books in simplest form.

All books: 6 + 9 = _____.

Math first: 6 : _____.

Divide both terms by _____: 2 : 5.

Answer: Math to all books = _____ : 5

You try

1. A pack has 10 pens and 15 pencils. Write each ratio in simplest form.
a. pens to pencils _____
b. pencils to all _____

_____

2. A team has 7 forwards and 4 defenders. Write the ratio of defenders to forwards.

_____

Sum it up: In a ratio, order matters because ___.

_____

Session 2 · Ratios and Tape Diagrams

I can use a tape diagram to share a total according to a given ratio. · 6.RP.A.1

Word / Spanish	Meaning — complete the blank	Example
tape diagram / diagrama de cinta	A drawing with _____ boxes that shows a ratio.	2 boxes : 3 boxes
unit (one box) / unidad	The amount inside _____ box.	24 ÷ 4 boxes = 6 in each box
total parts / partes totales	All the boxes _____ together.	3 + 1 = 4 parts

Key idea

Add the ratio terms to find the _____ parts.

One box = total ÷ _____.

Multiply one box by each _____ to find each share.

Worked example

Ana and Ben share 28 stickers in the ratio 3 : 4. Find each share.

Total parts: 3 + 4 = _____.

One box: 28 ÷ 7 = _____ stickers.

Ana: 3 × 4 = _____. Ben: 4 × 4 = 16.

Answer: Ana gets _____ and Ben gets _____. Check: 12 + 16 = 28

You try

1. Add the terms to find the total parts.
a. 3 : 5 _____
b. 7 : 2 _____
c. 4 : 6 _____

_____

2. Share $56 in the ratio 3 : 4. Find both shares.

_____

Sum it up: To share a total in a ratio, I first ___ because ___.

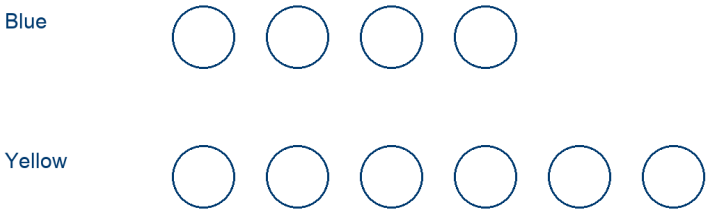
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Session 1 · Model, reason, and explain

Understanding Ratios · 6.RP.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A tray holds 4 blue counters and 6 yellow counters. Write blue:yellow, yellow:blue, and blue:all in simplest form. Explain what each number counts.



_____

_____

_____

Extend your reasoning

Add equal numbers of blue and yellow counters. Must the ratio stay the same? Give a counterexample.

_____

_____

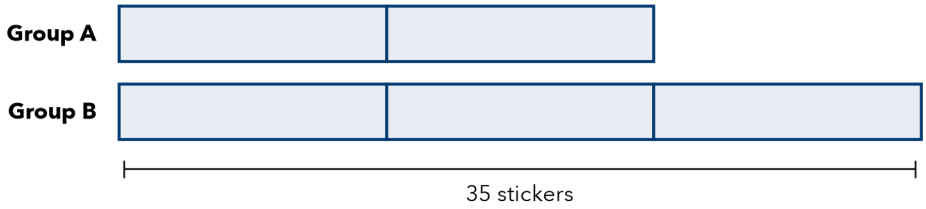
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Ratios and Tape Diagrams · 6.RP.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Split 35 stickers between two groups in the ratio 2:3. Label the equal tape units, find both shares, and check their sum.



_____

_____

_____

Extend your reasoning

Create a different total that can be shared in ratio 2:3 using whole stickers.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?